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AQA A Level Further Mathematics (7367)

Topic B: Complex Numbers

Sub topic B9: Exponential Form

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic B: Complex Numbers

Sub topic B9: Exponential Form

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

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Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

The function $f(x) = \cosh(\mathrm{i}x)$ is defined over the domain $\{x \in \mathbb{R}: -a\pi \le x \le a\pi\}$, where a is a positive integer.

By considering the graph of $y = [f(x)]^n$, find the mean value of $[f(x)]^n$, when n is an odd positive integer.

Fully justify your answer.

(Total 3 marks)

Q2.

(a) Show that $(1-\frac{1}{4}e^{2i\theta})(1-\frac{1}{4}e^{-2i\theta})=\frac{1}{16}(17-8\cos 2\theta)$

(3)

(b) Given that the series $e^{2i\theta}+\frac{1}{4}e^{4i\theta}+\frac{1}{16}e+\frac{1}{64}e^{8i\theta}+\dots$ has a sum to infinity, express this sum to infinity in terms of $e^{2i\theta}$

(2)

(c) Hence show that $\sum_{n=1}^{\infty}\frac{1}{4^{n-1}}\cos 2n\theta=\frac{16\cos 2\theta-4}{17-8\cos 2\theta}$

(4)

(d) Deduce a similar expression for $\sum_{n=1}^{\infty}\frac{1}{4^{n-1}}\sin 2n\theta$

(1)

(Total 10 marks)

Q3.

(a) Express $-4+4\sqrt{3}\,\mathrm{i}$ in the form $re^{i\theta}$, where r > 0 and $-\pi < \theta \le \pi$.

(3)

(b) (i) Solve the equation $z^3 = -4+4\sqrt{3}\,\mathrm{i}$, giving your answers in the form $re^{i\theta}$, where r > 0 and $-\pi < \theta \le \pi$.

(4)

(ii) The roots of the equation $z^3 = -4+4\sqrt{3}\,\mathrm{i}$ are represented by the points P, Q and R on an Argand diagram.

Find the area of the triangle PQR, giving your answer in the form $k\sqrt{3}$, where k is an integer.

(3)

(c) By considering the roots of the equation $z^3 = -4+4\sqrt{3}\,\mathrm{i}$, show that

$$\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} + \cos\frac{8\pi}{9} = 0$$

(4)
(Total 14 marks)

Q4.

(a) Write down the five roots of the equation $z^5 = 1$, giving your answers in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$.

(1)

(b) Hence find the four linear factors of

$$z^4 + z^3 + z^2 + z + 1$$

(3)

(c) Deduce that

$$z^2 + z + 1 + z^{-1} + z^{-2} = \left(z - 2\cos\frac{2\pi}{5} + z^{-1}\right)\left(z - 2\cos\frac{4\pi}{5} + z^{-1}\right)$$

(4)

(d) Use the substitution $z + z^{-1} = w$ to show that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

(6)
(Total 14 marks)

Q5.

(a) Express in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$:

(i) $4(1 + i\sqrt{3})$;

(ii) $4(1 - i\sqrt{3})$.

(3)

(b) The complex number z satisfies the equation

$$(z^3 - 4)^2 = -48$$

Show that $z^3 = 4 \pm 4\sqrt{3}i$.

(2)

(c) (i) Solve the equation

$$(z^3 - 4)^2 = -48$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(5)

(ii) Illustrate the roots on an Argand diagram.

(3)

(d) (i) Explain why the sum of the roots of the equation

$$(z^3 - 4)^2 = -48$$

is zero.

(1)

(ii) Deduce that $\cos\frac{\pi}{9} + \cos\frac{3\pi}{9} + \cos\frac{5\pi}{9} + \cos\frac{7\pi}{9} = \frac{1}{2}$.

(3)

(Total 17 marks)

Q6.

The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

where p and q are real, has roots α, β and γ.

It is given that $\alpha = 2 + 2\sqrt{3}i$.

(a) (i) Write down another root, β, of the equation.

(1)

(ii) Find the third root, γ.

(3)

(iii) Find the values of p and q.

(3)

(b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$.

(2)

(ii) Show that

$$(2 + 2\sqrt{3}i)^n = 4^n\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)$$

(2)

(iii) Show that

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1}\cos\frac{n\pi}{3} + \left(-\frac{1}{2}\right)^n$$

where n is an integer.

(3)

(Total 14 marks)

Q7.

(a) (i) Show that $\omega = e^{\frac{2\pi i}{7}}$ is a root of the equation $z^7 = 1$.

(1)

(ii) Write down the five other non-real roots in terms of ω.

(2)

(b) Show that

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$$

(2)

(c) Show that:

(i) $\omega^2 + \omega^5 = 2\cos\frac{4\pi}{7}$;

(3)

(ii) $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}$.

(4)

(Total 12 marks)

Q8.

(a) (i) Express each of the numbers $1 + \sqrt{3}\,i$ and $1 - i$ in the form $re^{i\theta}$, where $r > 0$.

(3)

(ii) Hence express

$$(1 + \sqrt{3}\,i)^8(1 - i)^5$$

in the form $re^{i\theta}$, where $r > 0$.

(3)

(b) Solve the equation

$$z^3 = (1 + \sqrt{3}\,i)^8\,(1 - i)^5$$

giving your answers in the form $a\sqrt{2}\,e^{i\theta}$, where a is a positive integer and $-\pi < \theta \le \pi$.

(4)

(Total 10 marks)

Q9.

(a) Show that

$$(z^4 - e^{i\theta})\,(z^4 - e^{-i\theta}) = z^8 - 2z^4\cos\theta + 1$$

(2)

(b) Hence solve the equation

$$z^8 - z^4 + 1 = 0$$

giving your answers in the form $e^{i\phi}$, where $-\pi < \phi \le \pi$.

(6)

(c) Indicate the roots on an Argand diagram.

(3)
(Total 11 marks)

Q10.

Given that $z = 2e^{\frac{\pi i}{12}}$ satisfies the equation

$$z^4 = a(1 + \sqrt{3}\,i)$$

where a is real:

(a) find the value of a;

(3)

(b) find the other three roots of this equation, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$.

(5)
(Total 8 marks)

Q11.

(a) Express $4 + 4i$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$.

(3)

(b) Solve the equation

$$z^5 = 4 + 4i$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \le \pi$.

(5)
(Total 8 marks)

Q12.

(a) Find the three roots of $z^3 = 1$, giving the non-real roots in the form $e^{i\theta}$, where $-\pi < \theta \le \pi$.

(2)

(b) Given that ω is one of the non-real roots of $z^3 = 1$, show that

$1 + \omega + \omega^2 = 0$

(2)

(c) By using the result in part (b), or otherwise, show that:

(i) $\frac{\omega}{\omega+1} = -\frac{1}{\omega}$;

(2)

(ii) $\frac{\omega^2}{\omega^2+1} = -\omega$;

(1)

(iii) $\left(\frac{\omega}{\omega+1}\right)^k + \left(\frac{\omega^2}{\omega^2+1}\right)^k = (-1)^k\, 2\cos\frac{2}{3}k\pi$, where k is an integer.

(5)

(Total 12 marks)

Q13.

(a) (i) Given that $z^6 - 4z^3 + 8 = 0$, show that $z^3 = 2 \pm 2i$.

(2)

(ii) Hence solve the equation

$z^6 - 4z^3 + 8 = 0$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(6)

(b) Show that, for any real values of k and θ,

$(z - ke^{i\theta})(z - ke^{-i\theta}) = z^2 - 2kz\cos\theta + k^2$

(2)

(c) Express $z^6 - 4z^3 + 8$ as the product of three quadratic factors with real coefficients.

(3)

(Total 13 marks)

Q14.

The complex numbers z_1 and z_2 are given by

$z_1 = \frac{1+i}{1-i}$ and $z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

(a) Show that $z_1 = i$.

(2)

(b) Show that $|z_1| = |z_2|$.

(2)

(c) Express both z_1 and z_2 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(3)

(d) Draw an Argand diagram to show the points representing z_1, z_2 and $z_1 + z_2$.

(2)

(e) Use your Argand diagram to show that

$$\tan \frac{5}{12}\pi = 2 + \sqrt{3}$$

(3)
(Total 12 marks)

Q15.

(a) Find the six roots of the equation $z^6 = 1$, giving your answers in the form $e^{i\phi}$, where $-\pi < \phi \leq \pi$.

(3)

(b) It is given that $w = e^{i\theta}$, where $\theta \neq n\pi$.

(i) Show that $\frac{w^2 - 1}{w} = 2i \sin \theta$.

(2)

(ii) Show that $\frac{w}{w^2 - 1} = -\frac{i}{2 \sin \theta}$.

(2)

(iii) Show that $\frac{2i}{w^2 - 1} = \cot \theta - i$.

(3)

(iv) Given that $z = \cot \theta - i$, show that $z + 2i = zw^2$.

(2)

(c) (i) Explain why the equation

$$(z + 2i)^6 = z^6$$

has five roots.

(1)

(ii) Find the five roots of the equation

$$(z + 2i)^6 = z^6$$

giving your answers in the form $a + ib$.

(4)
(Total 17 marks)